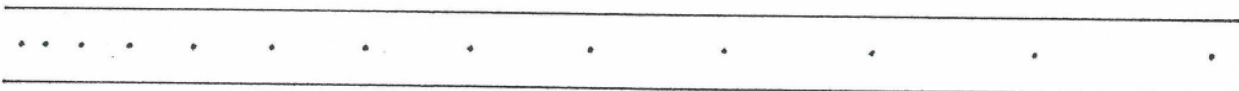


YEAR 10 SCIENCE (EXTENSION)
PHYSICAL SCIENCES
REVISION

1. (a) Distinguish between the following terms.
 - (i) Distance and displacement.
 - (ii) Speed and velocity.
 - (b) A man drives his car 8.00×10^2 m due East for 2.10 mins and then turns North, driving 3.00×10^2 m in 1.35 mins before stopping at a petrol station.
 - (i) Calculate the average speed for the entire motion.
 - (ii) Calculate the displacement made from home to the petrol station. Show your working clearly.
 - (iii) Determine the average velocity for the entire motion.
2. A group of students produced the following ticker tape. One dot is recorded every 0.02 seconds.
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- (a) Describe the type of motion shown by the tape.
 - (b) Analyse the tape and calculate a value for the motion shown. Show your working.
3. A car moves off from a standing start with a uniform acceleration. It reaches 20.0 ms^{-1} after 8.00 s before maintaining its speed for another 15.0 s. As it approaches a set of lights, it brakes uniformly to 8.00 ms^{-1} over 12.0 s. It maintains this velocity as it moves through the lights, which turned green as it approached..
 (Assume the movement has occurred on a straight road.)
- (a) Draw a velocity - time graph for the entire motion.
 - (b) By finding the gradient, calculate:
 - (i) the acceleration at the start of the motion.
 - (ii) the deceleration as the car approached the second set of lights.
 - (c) From the graph, determine the displacement of the car over 40.0 s of the motion.
4. A balloon filled with hydrogen gas has a constant velocity of 40.0 ms^{-1} East. How long will it take to go 4.50×10^2 m ?

5. A car of mass 1.53×10^3 kg is travelling at 70.0 kmh^{-1} along Hepburn Avenue when the driver has to brake sharply to avoid a small child who suddenly appeared on a bike crossing the road. The car slowed to 20.0 kmh^{-1} in 3.20 s and managed to miss the child, who made it across safely.
- Calculate the average force exerted by the brakes.
 - How far did the car travel while braking?
 - The driver of the car was an engineer and had a hard-hat sitting on the back shelf. Describe what happened to the hard-hat in this situation and explain why it took place.
6.
 - Explain the difference between **mass** and **weight**. Give an example to help your explanation.
 - The acceleration due to gravity on Jupiter is 22.5 ms^{-2} . Calculate the weight of a 4.50×10^2 kg object on Jupiter.
 - A Canadian astronaut is working on the International Space Station. He is able to float freely about the cabin, moving easily about with little pushes off the walls. During a link-up with students from an Australian school, one student asked him what it felt like to be weightless. The student was surprised when he replied "I'm not weightless so I can't tell you". Was he correct to say this? Explain your answer.
7. A 70.0 kg cyclist is applying a force of 1.50×10^2 N forwards but is working against a wind producing a friction force of 80.0 N opposing the movement.
- Calculate the acceleration of the cyclist.
 - It is possible for the rider to move at constant velocity. In terms of the forces acting, explain how this is possible.
8. You are sitting on a chair at the moment. Explain why you don't fall through the chair and are supported by it.
9. Write the following in **scientific notation** and with **standard units (kg, m, s)**.
- | | | |
|--------------------|-----------------------------|------------------|
| (a) 0.0293 mm | (b) 3652 kg | (c) 38.6 minutes |
| (d) 495 million km | (e) 70.0 kmh^{-1} | (f) 75 mg |
10. Perform the following calculation, expressing the answer to the correct number of significant figures.
- $$\frac{(21.5 + 356.2)}{(1.19 \times 10^{-4})} = \frac{1}{2}(340)v^2$$
11.
 - What is a **scalar** and a **vector**?
 - Give **two** examples of each.

12. During a stunt for a film, a 90.0 kg person leaps from a plane at 5,000 m above the ground. He maintains a streamlined body shape to minimise air resistance. Ignoring any friction, calculate the following.
- (a) How far has the person fallen after 18.0 s?
 - (b) How fast is he travelling at this point?
 - (c) The stunt person deploys a parachute that reduces his velocity to 25.0 ms^{-1} in 90.0 m. Calculate the force exerted by the parachute.
(Hint - calculate acceleration first.)
13. An electron of mass $9.11 \times 10^{-31} \text{ kg}$ is accelerated from rest to $2.45 \times 10^7 \text{ ms}^{-1}$ over a distance of 3.00 cm in an electric field.
- (a) Calculate the magnitude of the acceleration of the electron. Does this surprise you? Comment on this.
 - (b) Consider the size of the force required to accelerate the electron. Are you surprised by its magnitude? Comment on this.